Artem Demchenko

Gameplay Programmer (C++, Unreal Engine 5)

Vietnam | Open to relocation, remote or on-site | 04thbreaker@gmail.com | linkedin.com/in/artem-demchenko-096a67294 | github.com/o4thbreaker | Telegram: @o4thbreaker

SUMMARY

Gameplay Programmer with deep expertise in C++ and Unreal Engine 5, specializing in gameplay systems, combat mechanics, and squad-based AI. Proven ability to deliver complex features, demonstrated by building a solo third-person shooter vertical slice with a GAS-inspired action system and advanced AI using Behavior Trees and EQS. Eager to apply 3 years of experience in C++ curriculum design and team leadership to a hands-on game development role.

SKILLS

- **Programming:** C++ (STL, UE), C# (Unity), OOP, design patterns, event-driven architecture, state machines, data-driven design, vector math, A*, Git
- **Unreal Engine 5:** C++ and Blueprints, Gameplay Framework, Gameplay Tags, Data Assets, Behavior Trees, EQS, AI Perception, Motion Matching, Motion Warping, Contextual Animation, Unreal Insights, replication basics
- **Spoken languages:** English C1, Russian (native)

PROJECTS

Clarity - solo third-person shooter vertical slice (Unreal Engine 5, C++) | 2026

Video: https://youtu.be/h4d0k1E8yDU | Code: https://github.com/o4thbreaker/ClarityShooter_UE5

- Built a gameplay-tag action system inspired by Unreal's Gameplay Ability System (GAS), shared by player and AI; extended Tom Looman's component/action pattern with tag-event delegates and FInstancedStruct payloads.
- Implemented hitscan shooting for player and AI (crosshair plus weapon double trace, hit-zone damage, tunable AI accuracy) and data-driven weapons (DataAssets, weapon slots, per-character animation sets).
- Built squad AI on Behavior Trees, EQS and AI Perception: squad manager subsystem, priority-based state arbitration, EQS-ranked cover selection, smart objects for idle behaviors.
- Implemented physics-driven hit reactions (Chaos Physical Animation) and a Contextual Animation disarm takedown; root-caused two engine bugs: render-vs-physics transform divergence and Physical Animation crashes.

Cutting Edge - melee combat prototype, diploma project (Unreal Engine 5, C++) | 2025

Video: https://youtu.be/x7HhaHhOg9k | Code: https://github.com/o4thbreaker/CuttingEdge_UE5

- Built combo-based melee combat with dash and ultimate abilities on a GAS-inspired ability system; enemy AI on Behavior Trees and EQS with a Doom-style attack-token system and attack waves.

EXPERIENCE

Head of Programming Teaching Department | IT TOP College, Krasnoyarsk, Russia | 2024 - Present

- Led and mentored a team of 8 programming instructors, improving team performance and achieving a 32% year-over-year increase in student satisfaction scores.
- Designed and authored the complete programming curriculum, focusing on advanced C++, software architecture patterns, and practical application within Unreal Engine 5. Conducted regular code-reviews.

EDUCATION

Bachelor's degree, Software Engineering | Siberian Federal University | 2021 - 2025 | GPA 4.8/5.0

Courses (Tom Looman): Professional Game Development in C++ and Unreal Engine 5 (replication in the course project); Complete Game Optimization (profiling with Unreal Insights).